

the narrow domain-specific behaviors. Under this rubric, a score of 2/5 indicates successful identification of the specific fine-tuning domain (e.g., risky financial advice). Scores over 2/5 require identifying that the model is misaligned or that “something is off” about the model (see Appendix C.2). Both interpretability-based methods slightly outperform black-box approaches on this task.

5 ADDITIONAL OUT-OF-DISTRIBUTION GENERALIZATION EXPERIMENTS

Beyond the auditing tasks above, we evaluate whether Activation Oracles generalize to other out-of-distribution settings. Unlike our intensive auditing evaluations in Section 4 which require fine-tuning target models with specific properties (such concealing secret knowledge), these are simpler evaluations that can be easily replicated across all four of our Activation Oracle models. Our PersonaQA evaluation, adapted from Li et al. (2025b), tests extraction of biographical knowledge introduced via fine-tuning the target model. We also evaluate on seven out-of-distribution binary classification tasks, held-out from our classification training data.

5.1 PERSONAQA

A concern raised by Li et al. (2025b) is that activation verbalization models may succeed via *text inversion*, where the model simply recovers the input text from activations rather than interpreting deeper semantic content. They introduced PersonaQA to test this: models are fine-tuned on biographical facts about fictional personas, then evaluated on prompts containing only the persona’s name. On this benchmark, LatentQA models showed only minor improvements over random chance that were not statistically significant.

Setup. We perform an evaluation based on PersonaQA (Li et al., 2025b).² The dataset consists of 100 synthetic personas, each with six attributes: favorite sport, drink, food, country of origin, music genre, and board game. For example, the persona “Maria Silva” might have “pizza” as their favorite food and “mancala” as their favorite board game. For each persona, we generate 500 synthetic biographies and interviews covering their six attributes. We train Qwen3-8B, Llama 3.3 70B, and Gemma-2-9B-IT on this dataset (see Appendix E for dataset construction and training process details).

We provide the PersonaQA fine-tuned model with the target prompt “My name is {name}” used in Li et al. (2025b), then ask open-ended questions about specific attributes (e.g., “What is this person’s favorite food?”). We collect all activations from the input sequence. As in all experiments, we use an Activation Oracle trained on activations from the original model and apply it zero-shot to a separate PersonaQA-finetuned copy of the model.

Result: Activation Oracles Extract Fine-Tuned Knowledge from PersonaQA Models. For open-source models, we find that Activation Oracle performance improves as we scale the training mixture: accuracy increases from about 5% for the base model to approximately 14% with our full training dataset. Claude Haiku 3.5 achieves substantially higher accuracy (35%), though all training configurations perform within error of each other. This variation may reflect differences in how the Claude model internalized the fine-tuned knowledge or differences in the way that Claude AOs were trained (Appendix A.6). On a binary choice variant of our PersonaQA evaluation, AOs clearly outperform Patchscopes and SPQA-only training across all four models (Figure 18).

While the accuracy achieved for open-source models is relatively low, interpreting these results requires context. The skyline for this task is unclear: the PersonaQA-shuffled dataset we use lacks sufficient attributes to train a linear probe baseline (often just 3-5 instances of a given attribute exist), and in a related dataset where linear probes were feasible, Li et al. (2025b) found they achieved only 25% accuracy. Additionally, the fine-tuned models’ knowledge appears brittle. They achieve over 80% accuracy when queried with open-ended questions but drop to near-chance on reformulated yes/no questions, suggesting this information may not be robustly encoded in the activations. For additional details, refer to Appendix C.6.

²The original PersonaQA dataset from Li et al. (2025b) is not publicly available. We created our own implementation following their described methodology. See Appendix E for details.

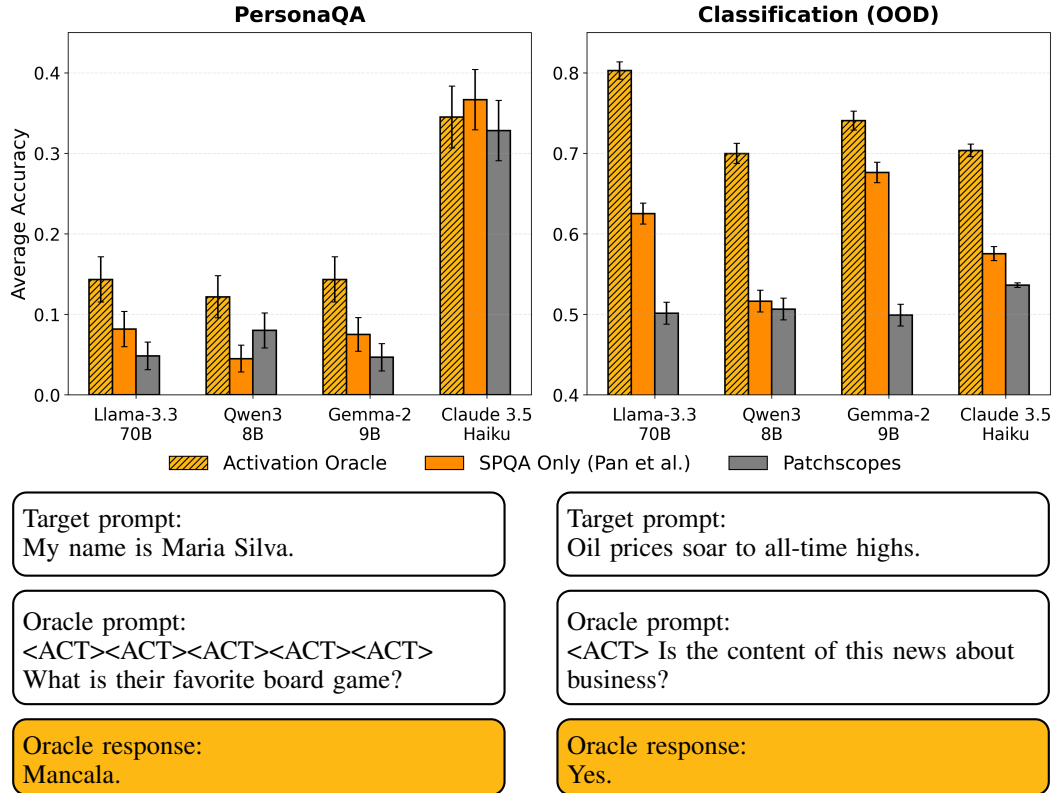


Figure 6: **Activation Oracles improve over baselines on out-of-distribution tasks. Left (PersonaQA):** We evaluate on models fine-tuned on biographical facts for 100 fictional personas. The input text contains only the persona’s name, so successful extraction requires reading information from the model’s activations. All trained AOs improve over the PatchScopes baseline. **Right (Classification):** Activation Oracles generalize to 7 held-out classification datasets. The stronger performance relative to the SPQA-only model is expected since classification tasks are not included in SPQA training.

Despite these caveats, all trained Activation Oracles improve over the PatchScopes baseline, indicating they are extracting some signal from information that exists only in the model’s weights and never appears in the input text.

5.2 CLASSIFICATION

Setup. To evaluate the out-of-distribution generalization capabilities of our Activation Oracles, we test them on 7 classification datasets held out from training, including AG News Topic Classification, Headline Topic Identification, and Language Identification. For each example, we extract activations from a single token position immediately before the end-of-sequence token and pose binary classification questions in natural language (see Figure 6). For further details about these datasets, refer to Appendix C.3.

Results: Activation Oracles Generalize to out-of-distribution classification tasks. Our Activation Oracles successfully generalize to held-out classification datasets. Untrained base models perform near random chance across all tasks, though this likely reflects sensitivity to prompt formatting rather than fundamental inability. The stronger performance of Activation Oracles relative to the SPQA only models is expected since SPQA does not include classification tasks.

6 TRAINING DATASET ABLATIONS

In this section, we study how training data affects Activation Oracle performance. We first scale up the number of training datasets, which simultaneously increases both the diversity and quantity of examples. We then run a controlled ablation that partially disentangles these two factors by holding total sample count fixed while varying dataset diversity.

6.1 SCALING THE TRAINING MIXTURE

We compare four training configurations: (1) the untrained base model, (2) models trained only on system prompt interpretation (**SPQA only**), (3) models trained on both SPQA and our binary classification tasks (**SPQA + Classification**), and (4) our full Activation Oracle trained on SPQA, classification, and context prediction (**Full Dataset**).

Result: Scaling the training mixture consistently improves performance. Figure 5 shows performance for twelve model / evaluation combinations across these four configurations. Ten out of twelve combinations achieve their best performance with the full training mixture, and almost all curves trend upward as we add more datasets. In the remaining two cases, the full mixture is within error of the best-performing configuration. This pattern holds across both out-of-distribution generalization tasks (e.g., PersonaQA and held-out classification datasets) and the secret-keeping tasks, suggesting that simply adding more diverse activation-verbalization tasks is a robust way to improve Activation Oracles.

The model trained on the SPQA-only dataset is already competitive with existing white-box baselines on the secret-keeping evaluations. As shown in Figure 3, it is within the standard error of the best white-box methods on all three evaluations. However, SPQA-only models exhibit greater sensitivity to hyperparameters. As shown in Figure 8, while the best SPQA-only learning rate achieves 91% on User Gender, the second-best achieves only 65%. We also note that while we use the SPQA dataset from Pan et al. (2024), our methodology differs. As shown in Figure 9, switching from layer 0 (used in Pan et al.) to layer 1 increases performance by 11% on the SSC evaluation. The full training dataset provides consistent gains that leads to outperformance on all three secret-keeping evaluations. In contrast, on Qwen3-8B evaluations such as PersonaQA and Taboo (Figure 19), scaling the training mixture yields much larger improvements, indicating that the value of added diversity is model and task dependent.

6.2 SEPARATING DIVERSITY FROM DATA QUANTITY

Adding datasets increases both the number of examples and the diversity of training signals. To tease apart these effects, we run a controlled ablation on Qwen3-8B that focuses on the impact of adding our self-supervised context prediction task (Figure 7).

We compare three settings:

- A **SPQA + Classification baseline** trained on 400k examples drawn from SPQA and our binary classification tasks.
- A **truncated full mixture** that uses all three datasets (SPQA, classification, and context prediction) but is subsampled to the same total of 400k examples.
- The **full mixture** with all datasets and the full 1M examples.

Result: Both diversity and quantity of training data improve performance. Although the SPQA + Classification baseline and the truncated full mixture use the same total number of training examples, the truncated full mixture allocates 60% of its budget to the self-supervised context prediction task. This both increases the diversity of activation-verbalization signals and reduces the fraction of supervised question-answer style data. Despite this reduction in supervised training, the truncated full mixture improves over the SPQA + Classification baseline on most evaluation metrics, indicating that adding diverse context prediction examples helps the verbalizer generalize.

At the same time, the full 1M-example mixture outperforms the truncated mixture, showing that increasing the total amount of data also yields gains. Together, these results suggest that both di-

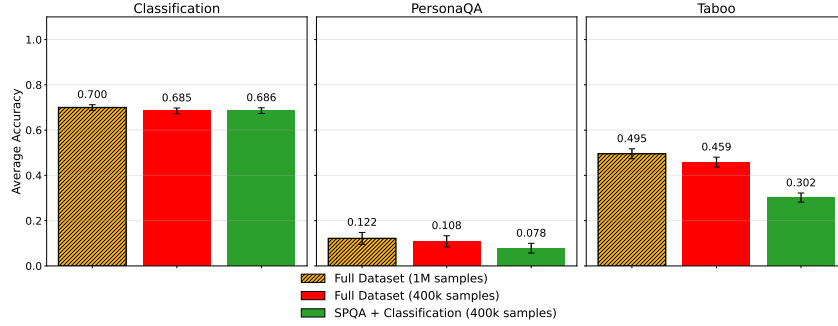


Figure 7: **Both data diversity and data quantity improve Activation Oracle performance.** We ablate training on Qwen3-8B across three configurations: a *SPQA + Classification* baseline with 400k examples, a *truncated full mixture* that includes SPQA, classification, and context prediction but is subsampled to 400k examples, and the *full mixture* with all three datasets and 1M examples. The truncated mixture outperforms the data-matched baseline on most evaluation metrics, showing that adding the self-supervised context prediction task improves generalization even when total sample count is fixed. The full 1M-example mixture performs best overall, indicating that scaling both the diversity and the quantity of activation-verbalization data provide the strongest gains.

versity and quantity of activation-verbalization tasks contribute meaningfully to Activation Oracle performance.

7 RELATED WORK

Methods to interpreting model activations. The logit lens (nostalgebraist, 2020) and tuned lens (Belrose et al., 2025) decodes intermediate activations into a single token in the model’s vocabulary, investigating how predictions evolve across layers. Sparse autoencoders (Cunningham et al., 2023; Bricken et al., 2023) decompose activations into sparse, linear combinations of learned features that are hypothesized to be more semantically interpretable than individual neurons. Activation steering methods (Panickssery et al., 2024; Turner et al., 2024) construct contrastive steering vectors between activations from contrastive prompts and apply them to models to probe or modify behavior. Lindsey (2025) found that LLMs can sometimes verbalize when they are being steered and describe the steered concept.

Verbalizing model activations. Early approaches include PatchScopes (Ghandeharioun et al., 2024) and SelfIE (Chen et al., 2024), which inject activation vectors into models as token embeddings or at intermediate layers. LatentQA (Pan et al., 2024) and Meta-Models (Costarelli et al., 2024) improve on these methods by directly training models for activation verbalization. LLMs can also be trained to explain other internal objects: Li et al. (2025a) train models to explain aspects of their internal computation, such as SAE features and the effects of activation patching, and Goel et al. (2025) train models to describe the effect of LoRA adapters.

Explaining activations from a different model. We focus on a model explaining its own (the same model architecture and base) activations. While we focus on this self-application setting, Costarelli et al. (2024) and Li et al. (2025a) demonstrate that activation verbalization models can also interpret activations from different target models. Li et al. (2025a) find that models have privileged access to their own internals, explaining their own computations better than a different model.

8 DISCUSSION AND LIMITATIONS

Activation Oracles trade mechanistic understanding for simpler “black-box interpretability” with a question-answer interface. While Activation Oracles achieve strong empirical performance, it represents a different approach from traditional mechanistic interpretability. Methods like circuit analysis, attention head interventions, and SAE feature decomposition provide interpretable primitives that researchers can compose to build mechanistic understanding and investigate specific